Leveraging offline experience to improve sample efficiency

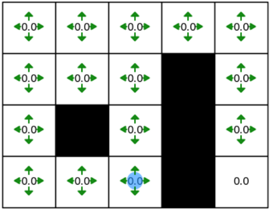
Olivier Sigaud

Sorbonne Université
http://people.isir.upmc.fr/sigaud

Outline

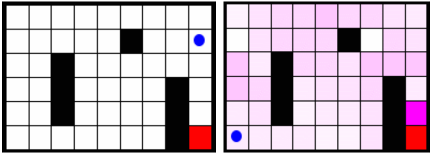
- Model-free RL is slow and sample inefficient
- This class: model-based methods and experience replay can improve learning efficiency
- Covered mechanisms:
  - Eligibility traces
  - Dyna variants: Dyna-PI, Dyna-Q, Dyna-AC, LCSs, Factored MDPs
  - Replay buffer
  - Prioritized sweeping, prioritized experience replay
  - Critical difference between experience replay and MBRL-based learning: imagined but unseen transitions
  - If time permits, some biology

Q-LEARNING in practice



- Build a states×actions table (*Q-Table*, eventually incremental)
- Initialise it (randomly or with 0 is not always the best choice)
- Apply $Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha[r_{t+1} + \gamma \max_{a \in A} Q(s_{t+1}, a) - Q(s_t, a_t)]$ after each action
- Problem: it is (very) slow

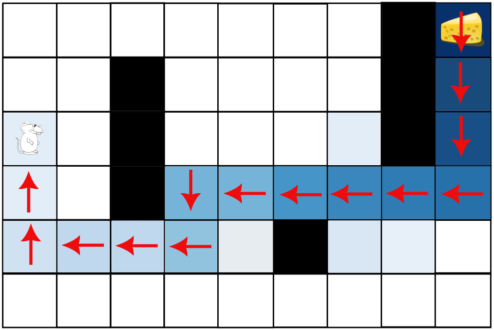
Model-based Reinforcement Learning: intuition



- Make profit of a learnt model of T and r

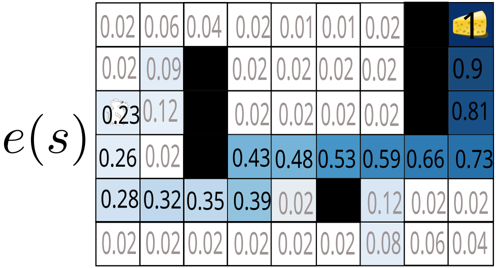
Eligibility traces

Eligibility traces: the replay idea



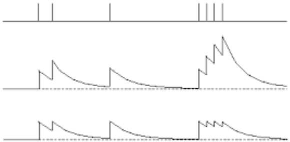
- Goal: improve over Q-LEARNING
- Naive approach: store all (s, a) pair along the trajectory and back-propagate values (along red arrows)
- Limited to finite horizon trajectories
- Converges faster, but needs more memory
- Trade-off between speed of convergence and memory

Eligibility traces: efficient implementation



- More sophisticated approach to deal with infinite horizon trajectories
- A variable $e(s)$ is set to $e(s) + 1$ (accumulating) or reinitialized to 1 (replacing) each time s is visited again and decayed with a factor λ after each time step
- $e_{t+1}(s) = \lambda e_t(s)$
- TD(λ): $\forall s \in S, V(s) \leftarrow V(s) + \alpha\delta e(s)$, (similar for SARSA ($\lambda$) and Q($\lambda$)),
- The most recently visited states are updated more
- Equivalent to back-up from trajectory, but memory in $|s|$, not in trajectory length

Eligibility traces: properties



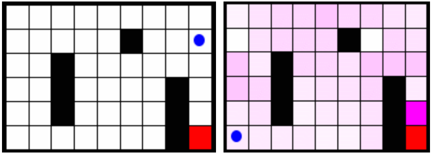
- TD(λ): $V(s) \leftarrow V(s) + \alpha\delta e(s)$, (similar for SARSA ($\lambda$) and Q($\lambda$)),
- $e_{t+1}(s) = \lambda e_t(s)$
- If $\lambda = 0$, $e(s)$ goes to 0 immediately
- Only the current transition is updated (no trace), thus we get TD($\lambda = 0$)=TD, SARSA $(\lambda = 0) =$ SARSA or Q-LEARNING $(\lambda = 0) =$ Q-LEARNING
- If $\lambda = 1$, $e(s)$ does not decay, we update all states equally and get Monte Carlo

Sutton, R. S. (1988) Learning to Predict by the Method of Temporal Differences. *Machine Learning*, 3:9–44

Singh, S. P. and Sutton, R. S. (1996) Reinforcement learning with replacing eligibility traces. *Machine learning*, 22:123–158

DYNA approaches

Model-based Reinforcement Learning: approaches



- Learning T and r is an incremental self-supervised learning problem
- With a model, an agent can play fictitious transitions "in the model"
- Two approaches:
  - First, learn a model, then learn a policy using the model
  - Learn a model and a policy simultaneously
- Question in lab: which is the most sample efficient?

Model acquisition first

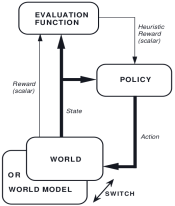
- Model acquisition questions:
  - How to collect data?: from a random policy, using an existing dataset, using an available policy
  - Which model to learn: stochastic or deterministic?
  - How to be sure the state×action space is covered?
  - When to stop?
- Policy learning questions:
  - Which algorithm to use?: policy iteration, Q-LEARNING, SARSA, ACTOR-CRITIC
  - When to stop?
- Once a good model is learnt, no need for more samples

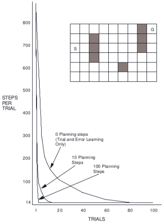
Simultaneous learning the model and the policy

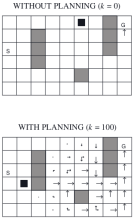
- Model acquisition is driven by the current policy
- If the policy is stuck, the model will not improve
- No "when to stop model acquisition?" question
- Similar questions: deterministic or stochastic, which algorithm?
- Additional question: how many updates of the policy per updates in the model?
- This approach defines the DYNA family

Sutton, R. S. (1991) DYNA, an integrated architecture for learning, planning and reacting. *SIGART Bulletin*, 2:160–163

The Dyna-AHC family





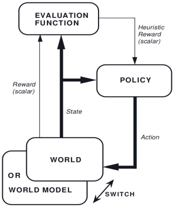


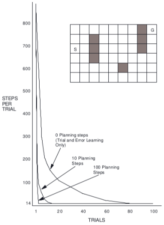
- AHC stands for Adaptive Heuristic Critic
- Interleaves back-ups in the model and in the environment
- Thanks to the model of transitions, DYNA can propagate values more often
- Fun fact: two nearly identical versions: ICML and N(eur)IPS

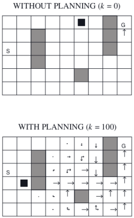
Sutton, R. S. (1990) Integrating architectures for learning, planning, and reacting based on approximating dynamic programming. In *Proceedings of the Seventh International Conference on Machine Learning*, pages 216–224, San Mateo, CA. Morgan Kaufmann

Sutton, R. S. (1990) Integrated modeling and control based on reinforcement learning and dynamic programming. *Advances in neural information processing systems*, 3

Dyna-PI, Dyna-Q, Dyna-AC

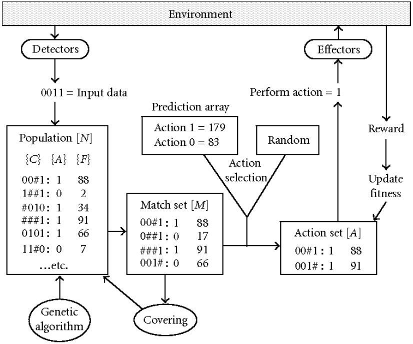






- Several approaches:
  - Draw random transitions in the model ("in imagination") and apply TD back-ups: DYNA-Q, DYNA-AC
  - Do the same along Monte Carlo trajectories in imagination: DYNA-PI
  - Better propagation: Prioritized Sweeping (see later)
  - Other point: model generalization

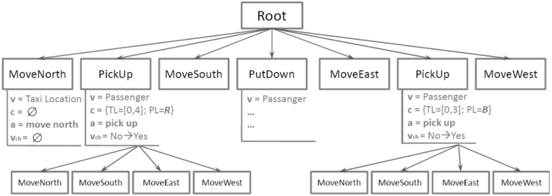
Generalization 1: Anticipatory Learning Classifier Systems



- Problem: in the stochastic case, the model of transitions is in $card(S) \times card(S) \times card(A)$
- Usefulness of compact models
- MACS: DYNA with generalisation
- Lots of heuristic processes

Gérard, P., Meyer, J.-A., & Sigaud, O. (2005) Combining latent learning with dynamic programming in MACS. *European Journal of Operational Research*, 160:614–637

Generalization 2: Factored MDPs



- SPITI: DYNA with generalisation (Factored MDPs)
- Much better founded formalization
- Could address very large MDPs
- A bot at CounterStrike with boolean features

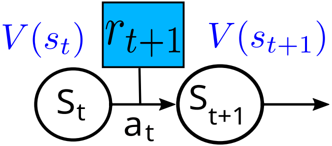
Degris, T., Sigaud, O., & Wuillemin, P.-H. (2006) Learning the Structure of Factored Markov Decision Processes in Reinforcement Learning Problems. *Proceedings of the 23rd International Conference on Machine Learning (ICML'2006)*, pages 257–264

Replay buffer

Introduction: Experience replay (ER) and replay buffer (RB)

- In MFRL, the agent learns from experience samples "as they come"
- Instead, it may store them and learn from them later (ER)
- The structure to store them is called a replay buffer (RB)
- Replay buffer is good for:
  - decorrelating samples: improving stability in Deep RL
  - reusing samples: improving sample efficiency
- Biological counterpart:
  - replayed experience = episodic memory
  - Replay buffer = hippocampus

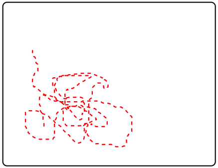
Replayed Samples



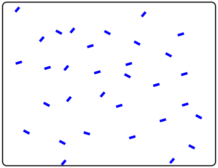
- Idea: instead of learning from transitions immediately, store them and learn asynchronously
- Samples must contain the necessary information to update the critic and/or actor
- In Q-LEARNING, $\{s_i, a_i, r_{i+1}, s_{i+1}\}$ is enough (most common case)
- In SARSA, one needs $\{s_i, a_i, r_{i+1}, s_{i+1}, a_{i+1}\}$
- NB: there is no difference between an algorithm drawing random transitions from a model and an algorithm drawing random transitions from a RB
- Key intuition: without generalization, MBRL and ER share the same properties

Vanseijen, H. and Sutton, R. (2015) A deeper look at planning as learning from replay. In *Proceedings of the 32nd International Conference on Machine Learning*, pages 2314–2322

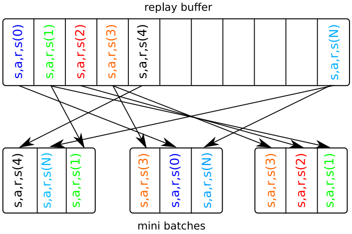
Replay buffer properties: Improving stability



Non i.i.d. samples



i.i.d. samples



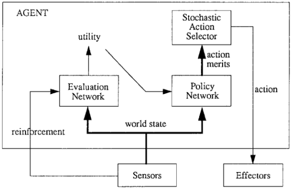
- Agent samples are not independent and identically distributed (i.i.d.)
- Shuffling a replay buffer (RB) makes them more i.i.d.
- Recent data in the RB come from policies close to the current one

de Bruin, T., Kober, J., Tuyls, K., & Babuška, R. (2015) The importance of experience replay database composition in deep reinforcement learning. In *Deep RL workshop at NIPS 2015*

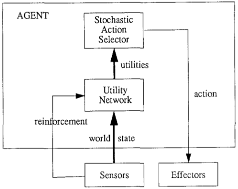
Replay buffer properties: Improving sample efficiency

- Reusing samples improves a lot the sample efficiency
- Important intuition: in the discrete deterministic case, a RB containing one sample from each (state, action) pair is enough for Q-LEARNING to converge
- In the stochastic case, RB samples should reflect the distribution over the next state
- This may require a large RB (over 1 million samples)
- In the continuous case, the state (and action) spaces cannot be covered
- But deep RL algorithms using a RB still benefit from sample reuse

Historical pioneer: Long J. Lin



AHCON

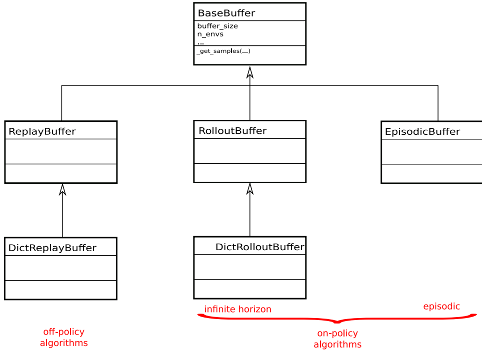


QCON

- AHC: Adaptive Heuristic Critic
- AHCON is a connexionist AHC (like DDPG)
- QCON is a connexionnist Q-LEARNING (like DQN)
- Proposes AHCON-R and QCON-R, "-R" for "with Replay" (of past experience)
- That's the naive idea behind eligibility traces
- Two other combinations:
  - AHCON-M and QCON-M for model-based
  - AHCON-T and QCON-T for teacher (imitation learning)
- Combines MBRL, experience replay and imitation

Lin, L.-J. (1992) Self-Improving Reactive Agents based on Reinforcement Learning, Planning and Teaching. *Machine Learning*, 8(3/4):293–321

Variety of RBs



- Different ways to store (one list, by episodes, ...)
- Different ways to retrieve (shuffled, prioritized, different mini-batch sizes)
- Different ways to forget (generally FIFO, no forget, ...)
- The effect of variants are still poorly understood

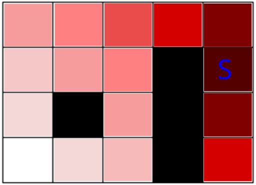
Zhang, S. & Sutton, R. S. (2017) A deeper look at experience replay. *arXiv preprint arXiv:1712.01275*

The successor representation

The successor representation (SR): several perspectives

- Another way to perform RL
  - Quick replanning when the MDP changes but not the reward, or vice versa
  - An intermediate between MFRL and MBRL
  - An alternative to their combination
- A computational tool in prioritized replay
  - A way to determine states that will be visited soon

The SR: definition



- A different way to compute the value function
- $V(s) = \sum_{s'} M_\pi(s, s')R(s')$
- The $M_\pi(s, s')$ matrix defines the discounted probability of being in s' when being in s before and following π
- Can be learned with a TD mechanism, where the reward is replaced by state identity
- $\delta_t = \mathbf{I}(s_t = s') + \gamma M_\pi(s_{t+1}, s') - M_\pi(s_t, s')$
- $M_\pi(s_t, s') \leftarrow M_\pi(s_t, s') + \alpha\delta_t$
- See `https://bair.berkeley.edu/blog/2021/01/05/successor/`

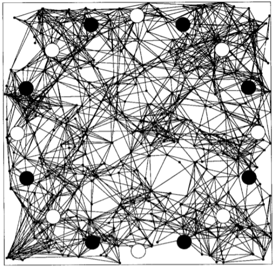
Dayan, P. (1993) Improving generalization for temporal difference learning: The successor representation. *Neural Computation*, 5(4):613–624

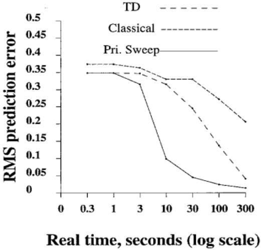
Prioritized sweeping
vs
prioritized experience replay

Prioritized reuse of experience: two frameworks

- In default experience reuse algorithms, samples are selected randomly
- This can be inefficient
- Question: how to improve the efficiency of experience reuse?
  - Prioritized Sweeping: In a DYNA architecture, when learning "in imagination", instead of generating random transitions, one might generate the most useful ones
  - Prioritized Experience Replay: In MFRL, when using a RB, instead of drawing random samples, one might select the most useful ones
- They were two simultaneous prioritized sweeping papers:
  - Prioritized Sweeping (Moore&Atkeson)
  - Queue-Dyna (Peng&Williams)

Prioritized Sweeping

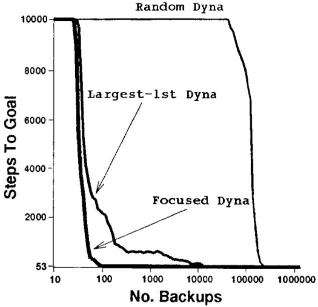




- They use a DYNA architecture, this is MBRL
- What are the most useful transitions?
- In Q-LEARNING, when discovering reward, the value is not backpropagated
- This is a waste of time (use eligibility)
- Update the transition whose TD error is the largest (recent past)
- Only backward replay: replay the samples with the highest TD error

Moore, A. W. & Atkeson, C. (1993). Prioritized sweeping: Reinforcement learning with less data and less real time. *Machine Learning*, 13:103–130.

Queue-Dyna



- They call it DYNA-style, but they just add replay to Q-LEARNING
- What are the most useful transitions?
  - Backward replay: replay the samples with the highest TD error (recent past)
  - Forward replay: replay the samples most likely to be visited soon
  - Focused DYNA: naive version of the SR (the term was not coined)
  - Strong link to biology: Mattar&Daw

Peng, J. & Williams, R. J. (1993) Efficient learning and planning within the dyna framework. *Adaptive behavior*, 1(4):437–454

Mattar, M. G. and Daw, N. D. (2018) Prioritized memory access explains planning and hippocampal replay. *Nature neuroscience*,

SR: three ways to compute it

- In Dayan, the TD-based way
- In Peng and Williams:
  - $Need(s_t) = \gamma^{d(s_t)}[r_t + \gamma V(s_{t+1})]$ where
  - $d(s_t)$ is the min distance from the start state to state s_t
- In Mattar and Daw:
  - $Need(s_t) = (Id - \gamma T)^{-1}[s_t]$ where
  - Id is the identity matrix, T is the transition matrix (with which actions?) and $[s_t]$ is the s_t^{th} line of the matrix
- The Peng and Williams' way only works in the deterministic case where the reward is 1 at the goal and 0 everywhere else...
- Same year as Dayan's SR paper

Prioritized Experience Replay

S_0	S_1	S_2	S_3	S_4	S_5	S_6	S_7	... ...
$\delta = 0.8$	$\delta = 0.7$	$\delta = 0.5$	$\delta = 0.2$	$\delta = 0.1$	...	...	$\delta = 0.01$	

- In the context of DQN (deep reinforcement learning)
- Samples with a greater TD error improve the critic faster
- Give them a higher probability of being selected
- Favors the replay of new (s, a) pairs (largest TD error)
- Several minor hacks, and interesting discussion
- Converges twice faster than DQN

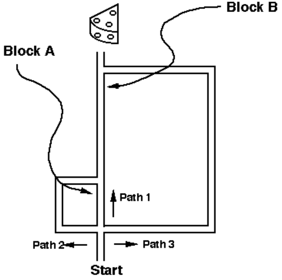
Schaul, T., Quan, J., Antonoglou, I., & Silver, D. (2015) Prioritized Experience Replay. *arXiv preprint arXiv:1511.05952*

Difference between MBRL transitions and RB samples

- The RB case:
  - In experience replay, samples are drawn from a replay buffer,
  - Sampled transitions must have already been seen by the agent
- The MBRL case:
  - If the model is tabular, only seen transitions have non zero probability
  - Sampled transitions also must have already been seen by the agent
  - But if the model is a function approximator...
  - ... it may generalize and generate transitions that have never been seen
- By contrast to experience replay based agents, (deep) MBRL agent may train on samples they have not seen
- To my knowledge, no prioritized transition sampling in (deep) MBRL
- Is there evidence of this in biological agents?

Biological counterpart
MBRL processes

Tolman's experiments (evidence for MBRL)



- The rat knows the maze, it uses the shortest path (Path 1) to reach food
- First, we block A. The rat uses Path 2 rather than Path 3
- If we block B, the rat goes straight to Path 3. It "knows" that blocking B also blocks Path 2.
- The rat builds a "cognitive map" (a model)

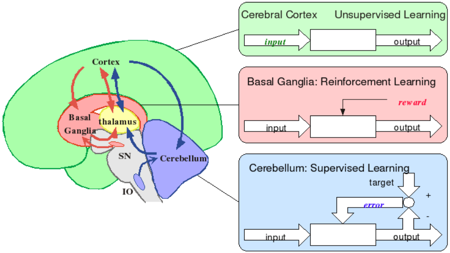
- Important starting point of cognitive sciences against behaviorism

Tolman, E. C. and Honzik, C. H. (1930) Introduction and removal of reward and maze performance in rats. *University of california Publications in Psychology*, 4:257–275

Tolman, E. C. (1948) Cognitive maps in rats and men. *Psychological Review*, 55:189–208

Martinet, L.-E., Passot, J.-B., Fouque, B., Meyer, J.-A., and Arleo, A. (2008) Map-based spatial navigation: A cortical column model for action planning. In *Spatial Cognition VI* pages 39–55. Springer

Learning with a brain



- Cerebellum: learning a predictive model of the world
- Basal Ganglia: reinforcement learning
- Cortex: representation learning
- The brain would be performing MBRL over efficient representations

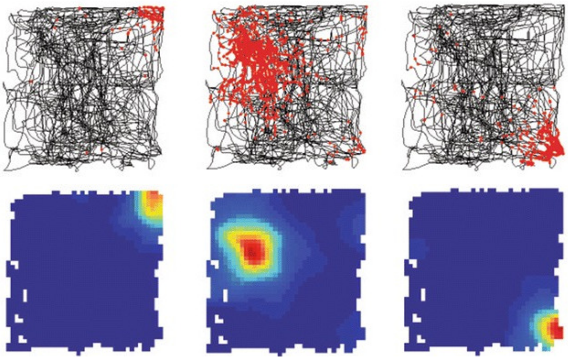
Doya, K. (2000) Complementary roles of basal ganglia and cerebellum in learning and motor control. *Current Opinion in Neurobiology*, 10:732–739

Biological counterpart
Experience replay

Deeper computational neuroscience models

- Navigation in rodents is an adequate biological paradigm:
  - Rodents locate themselves with “place cells” in the hippocampus
  - Place cells are easy to record, one see replays
  - With replays in place cells, one can see which places the rodent evoques
  - Reactivations take place before, during and after navigation, as well as during sleep
- This make it easy to study the corresponding “algorithms”

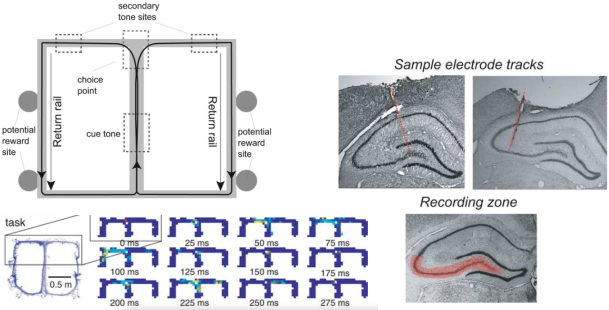
Place cells



- Rodents locate themselves with "place cells" in the hippocampus
- Each place activates a specific population of cells
- There are also grid cells, border cells, ...

Poucet, B., Lenck-Santini, P.-P., Paz-Villagrán, V., and Save, E. (2003) Place cells, neocortex and spatial navigation: a short review. *Journal of Physiology-Paris*, 97(4-6):537–546

Recording in place cells



- Place cells are located in the hippocampus
- Recording place cells activity tells the location the rodent "thinks about"

Johnson, A. and Redish, A. D. (2007) Neural ensembles in ca3 transiently encode paths forward of the animal at a decision point. *Journal of Neuroscience*, 27(45):12176–12189

Replay during navigation

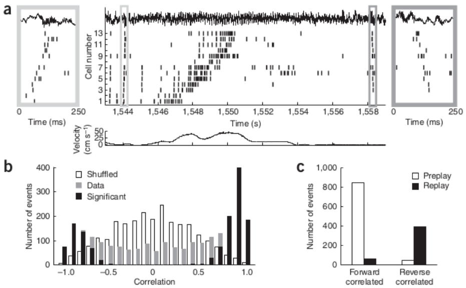
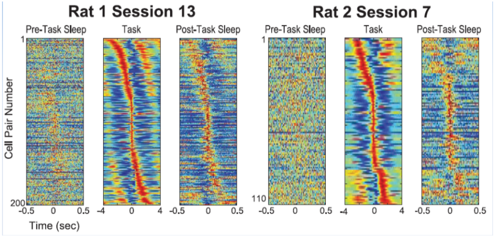


Figure 1 Forward and reverse pre- and replay of place-cell sequences. (**a**) Spike trains of 13 neurons before, during, and after a single lap (CA1 local field potential shown on top; velocity of the rat shown in the lower panel). The left and right insets magnify 250-ms sections of the spike train, depicting forward preplay and reverse replay, respectively. (**b**) Histograms show rank-order correlations of the immobility sequences (gray) and an equal number of shuffled surrogate events (white) to the place-field run sequence template. Significantly correlated events are in black. (**c**) 95% of significant forward correlated events took place before the run (preplay), whereas 85% of significant reverse correlated events took place after the run (replay).

- When recording during navigation, one can observe forward and backward replay in place cells

Diba, K. and Buzsáki, G. (2007) Forward and reverse hippocampal place-cell sequences during ripples. *Nature neuroscience*, 10(10):1241–1242

Reactivation during sleep

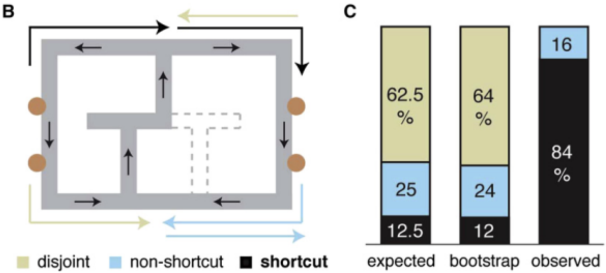


- When recording at night, one can see the rodent "dreaming" some parts of trajectories
- Importantly, frontal cortex activity correlates with hippocampus activity
- Suggests transfer of information from hippocampus to cortex

Wilson, M. A. and McNaughton, B. L. (1994) Reactivation of hippocampal ensemble memories during sleep. *Science*, 265(5172):676–679

Euston, D. R., Tatsuno, M., and McNaughton, B. L. (2007) Fast-forward playback of recent memory sequences in prefrontal cortex during sleep. *Science*, 318(5853):1147–1150

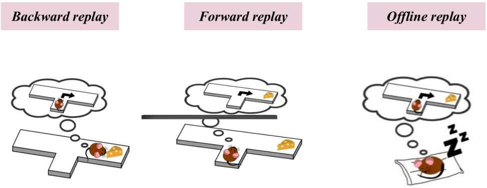
Complex relationship between replay and experience



- Replay samples more what is more informative, not in proportion to experience
- Replay without following trajectories can discover shorter paths

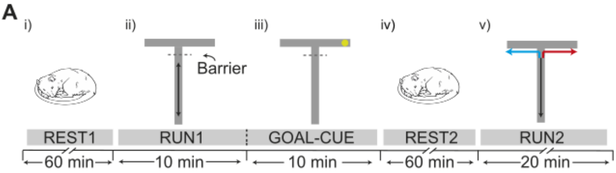
Gupta, A. S., Van Der Meer, M. A., Touretzky, D. S., and Redish, A. D. (2010) Hippocampal replay is not a simple function of experience. *Neuron*, 65(5):695–705

Prioritized replay in rats



- Mattar & Daw, Main objectives:
  - Find a criterion for sample selection that best explains biological replay
  - Assumption: replay is prioritized to maximize efficiency
  - Account for the emergence of backward replay and forward replay
  - Explain properties of offline replay
- Key abstraction step: no attempt to account for detailed biological experiments
- Rather try to reproduce qualitative findings

(Tentative) evidence in favor of MBRL



- The rat sees the environment (place cells can be set) but does not navigate in it (no experience to be replayed)
- The evidence for replay of unseen transitions is not really there

Ólafsdóttir, H. F., Barry, C., Saleem, A. B., Hassabis, D., and Spiers, H. J. (2015) Hippocampal place cells construct reward related sequences through unexplored space. *Elife*, 4:e06063

Conclusions, outstanding questions

Conclusions

- About the difference between experience replay and MBRL:
  - There is some confusion in the AI literature (Queue-Dyna)
  - Most comp. neuroscience studies only account for seen experience replay
  - Tabular representations do not come with generalization
  - To model MBRL-based replay, need for generalization to unseen places
- Before deep MBRL, no tool to model learning from unseen experience
- Better tools are now available
- Can we ask questions before having the tools?
- Progress in AI may induce progress in (computational) neuroscience

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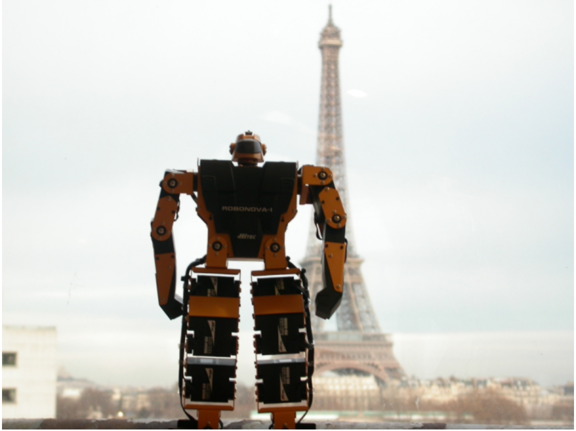
Understanding agents with multiple metrics

- Multidimensional analysis is key for truly understanding algorithm performance
- Include behavioural metrics and internal metrics, not just performance metrics
- Computational neuroscience papers do it much more than AI papers

Any question?



Send mail to: `Olivier.Sigaud@isir.upmc.fr`

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